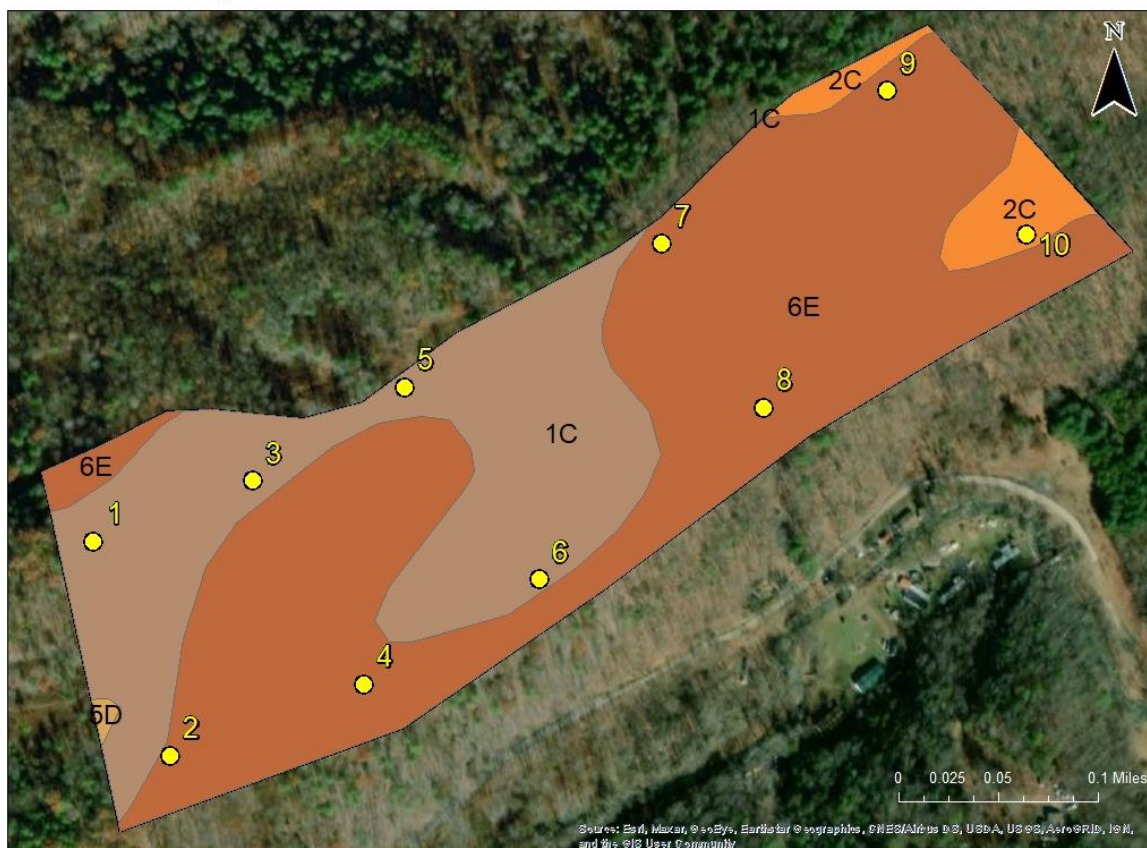


Virginia Tech Fishburn Enviornmental and Soil Report

Montgomery County, Virginia (VA121)			
Map Unit Symbol	Map Unit Name	Acres in AOI	Percent of AOI
1C	Berks-Clymer complex, 7 to 15 percent slopes	11.3	33.1%
2C	Berks-Groseclose complex, 7 to 15 percent slopes	1.4	4.2%
5D	Berks-Weikert complex, 15 to 25 percent slopes	0.1	0.2%
6E	Berks and Weikert soils, 25 to 65 percent slopes	21.4	62.6%
Totals for Area of Interest		34.3	100.0%

VT Fishburn - Soil Type



Montgomery County, Virginia

1C—Berks-Clymer complex, 7 to 15 percent slopes

- *National map unit symbol:* kc2c
- *Elevation:* 1,700 to 3,000 feet
- *Mean annual precipitation:* 30 to 45 inches
- *Mean annual air temperature:* 50 to 57 degrees F
 - *Frost-free period:* 117 to 185 days
- *Farmland classification:* Not prime farmland

Map Unit Composition

- *Berks and similar soils:* 50 percent
- *Clymer and similar soils:* 30 percent

Description of **Berks**

Setting

- *Landform:* Hills
- *Landform position (two-dimensional):* Backslope, summit
- *Landform position (three-dimensional):* Side slope, interfluve
- *Down-slope shape:* Convex
- *Across-slope shape:* Convex
- *Parent material:* Shale, siltstone, and sandstone residuum

Typical profile

- *H1 - 0 to 7 inches:* channery silt loam
- *H2 - 7 to 23 inches:* very channery silt loam
- *H3 - 23 to 33 inches:* extremely channery silt loam
- *H4 - 33 to 79 inches:* bedrock

Properties and qualities

- *Slope:* 7 to 15 percent
- *Depth to restrictive feature:* 20 to 40 inches to lithic bedrock
- *Drainage class:* Well drained
- *Runoff class:* Very high
- *Capacity of the most limiting layer to transmit water (Ksat):* Very low to high (0.00 to 5.95 in/hr)
- *Depth to water table:* More than 80 inches
- *Frequency of flooding:* None
- *Frequency of ponding:* None
- *Available water capacity:* Very low (about 2.5 inches)

Description of **Clymer**

Setting

- *Landform:* Hills
- *Landform position (two-dimensional):* Backslope, summit
- *Landform position (three-dimensional):* Side slope, interfluve
- *Down-slope shape:* Convex
- *Across-slope shape:* Convex
- *Parent material:* Sandstone, siltstone, and shale residuum

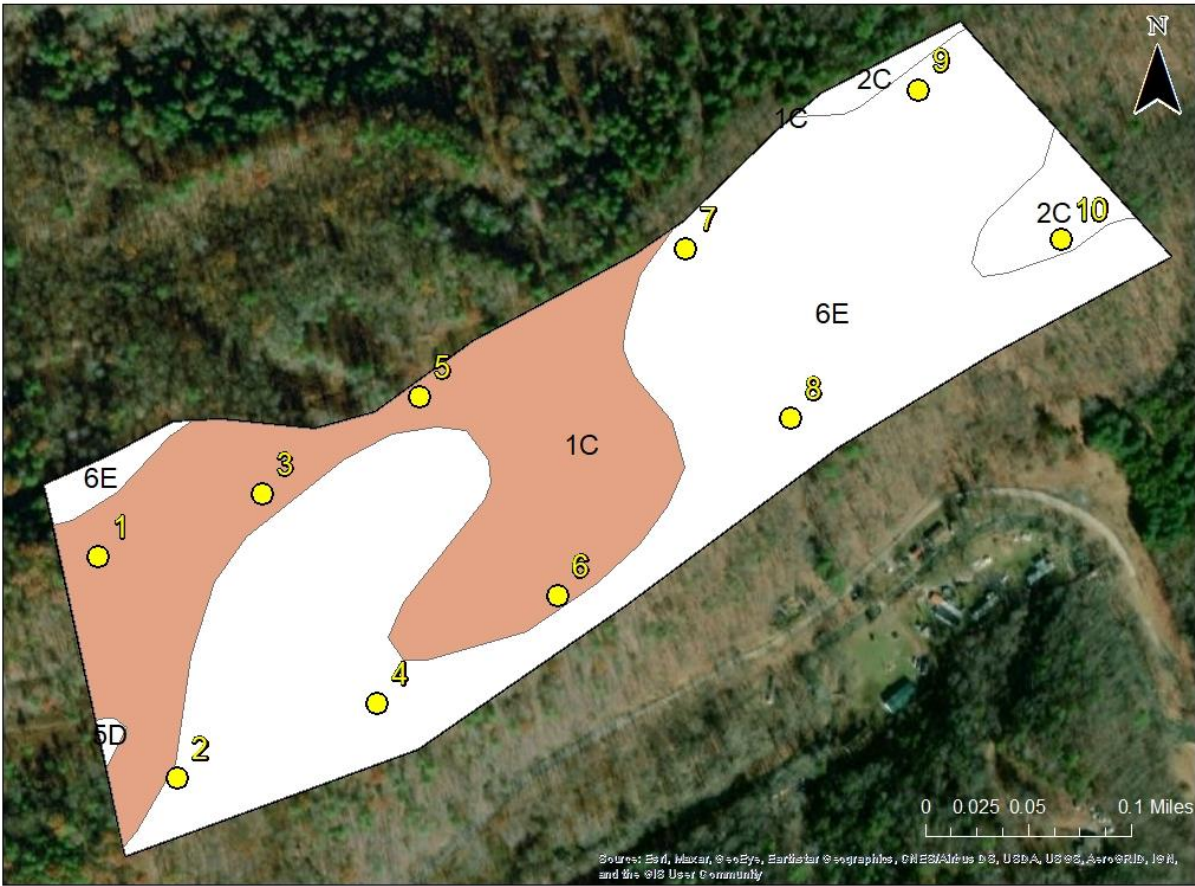
Typical profile

- *H1 - 0 to 9 inches:* loam
- *H2 - 9 to 21 inches:* clay loam
- *H3 - 21 to 32 inches:* channery sandy clay loam
- *H4 - 32 to 49 inches:* very channery sandy loam
- *H5 - 49 to 79 inches:* bedrock

Properties and qualities

- *Slope:* 7 to 15 percent
- *Depth to restrictive feature:* 40 to 60 inches to lithic bedrock
- *Drainage class:* Well drained
- *Runoff class:* Medium
- *Capacity of the most limiting layer to transmit water (Ksat):* Very low (0.00 to 0.00 in/hr)
- *Depth to water table:* More than 80 inches
- *Frequency of flooding:* None
- *Frequency of ponding:* None
- *Available water capacity:* Low

VT Fishburn - Soil Map - 1C



Montgomery County, Virginia

2C—Berks-Groseclose complex, 7 to 15 percent slopes

- *National map unit symbol:* kc2t
- *Elevation:* 1,700 to 3,000 feet
- *Mean annual precipitation:* 30 to 45 inches
- *Mean annual air temperature:* 50 to 57 degrees F
 - *Frost-free period:* 117 to 185 days
- *Farmland classification:* Not prime farmland

Map Unit Composition

- *Berks and similar soils:* 40 percent
- *Groseclose and similar soils:* 35 percent

Description of Berks

Setting

- *Landform:* Hills
- *Landform position (two-dimensional):* Backslope, summit
- *Landform position (three-dimensional):* Side slope, interfluvium
- *Down-slope shape:* Convex
- *Across-slope shape:* Convex
- *Parent material:* Shale, siltstone, and sandstone residuum

Typical profile

- *H1 - 0 to 7 inches:* channery silt loam
- *H2 - 7 to 23 inches:* very channery silt loam
- *H3 - 23 to 33 inches:* extremely channery silt loam
- *H4 - 33 to 79 inches:* bedrock

Properties and qualities

- *Slope:* 7 to 15 percent
- *Depth to restrictive feature:* 20 to 40 inches to lithic bedrock
- *Drainage class:* Well drained
- *Runoff class:* Very high
- *Capacity of the most limiting layer to transmit water (Ksat):* Very low to high (0.00 to 5.95 in/hr)
- *Depth to water table:* More than 80 inches
- *Frequency of flooding:* None
- *Frequency of ponding:* None
- *Available water capacity:* Very low (about 2.5 inches)

Description of Groseclose

Setting

- *Landform:* Hills
- *Landform position (two-dimensional):* Backslope, summit
- *Landform position (three-dimensional):* Side slope, interfluvium
- *Down-slope shape:* Convex
- *Across-slope shape:* Convex
- *Parent material:* Limestone, shale, siltstone, and sandstone residuum

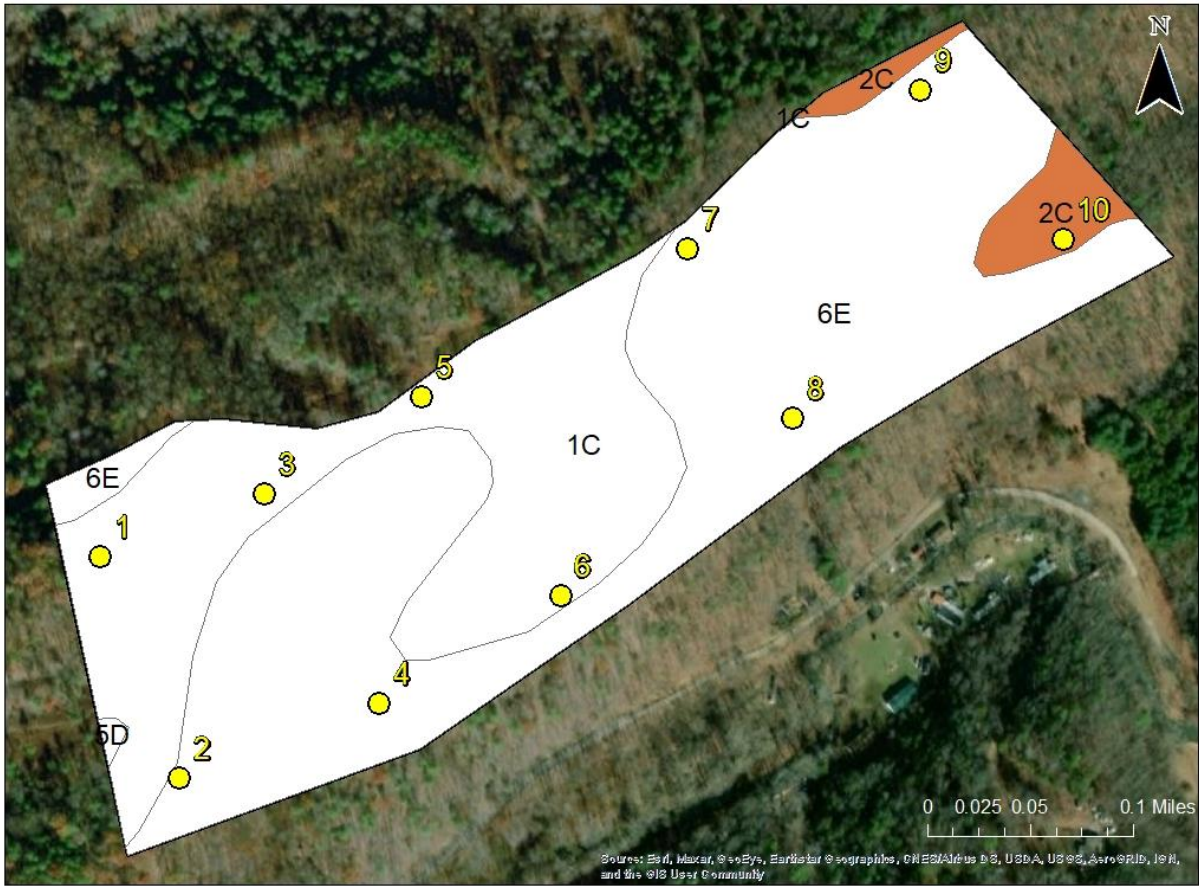
Typical profile

- *H1 - 0 to 6 inches:* silt loam
- *H2 - 6 to 28 inches:* clay
- *H3 - 28 to 39 inches:* clay
- *H4 - 39 to 51 inches:* clay
- *H5 - 51 to 79 inches:* clay loam

Properties and qualities

- *Slope:* 7 to 15 percent
- *Depth to restrictive feature:* More than 80 inches
- *Drainage class:* Well drained
- *Runoff class:* High
- *Capacity of the most limiting layer to transmit water (Ksat):* Moderately low to moderately high (0.06 to 0.20 in/hr)
- *Depth to water table:* More than 80 inches
- *Frequency of flooding:* None
- *Frequency of ponding:* None
- *Available water capacity:* Moderate (about 8.7 inches)

VT Fishburn - Soil Map - 2C



Montgomery County, Virginia

6E—Berks and Weikert soils, 25 to 65 percent slopes

- *National map unit symbol:* kc38
- *Elevation:* 1,700 to 3,000 feet
- *Mean annual precipitation:* 30 to 45 inches
- *Mean annual air temperature:* 50 to 57 degrees F
 - *Frost-free period:* 117 to 185 days
- *Farmland classification:* Not prime farmland
 - *Berks and similar soils:* 50 percent
 - *Weikert and similar soils:* 25 percent

Description of **Berks**

Setting

- *Landform:* Hills
- *Landform position (two-dimensional):* Backslope
- *Landform position (three-dimensional):* Side slope
- *Down-slope shape:* Linear
- *Across-slope shape:* Convex
- *Parent material:* Shale, siltstone, and sandstone residuum

Typical profile

- *H1 - 0 to 7 inches:* channery silt loam
- *H2 - 7 to 23 inches:* very channery silt loam
- *H3 - 23 to 33 inches:* extremely channery silt loam
- *H4 - 33 to 79 inches:* bedrock

Properties and qualities

- *Slope:* 25 to 65 percent
- *Depth to restrictive feature:* 20 to 40 inches to lithic bedrock
- *Drainage class:* Well drained
- *Runoff class:* Very high
- *Capacity of the most limiting layer to transmit water (Ksat):* Very low to high (0.00 to 5.95 in/hr)
- *Depth to water table:* More than 80 inches
- *Frequency of flooding:* None
- *Frequency of ponding:* None
- *Available water capacity:* Very low (about 2.5 inches)

Description of **Weikert**

Setting

- *Landform:* Hills
- *Landform position (two-dimensional):* Backslope
- *Landform position (three-dimensional):* Side slope
- *Down-slope shape:* Linear
- *Across-slope shape:* Convex
- *Parent material:* Shale, siltstone, and sandstone residuum

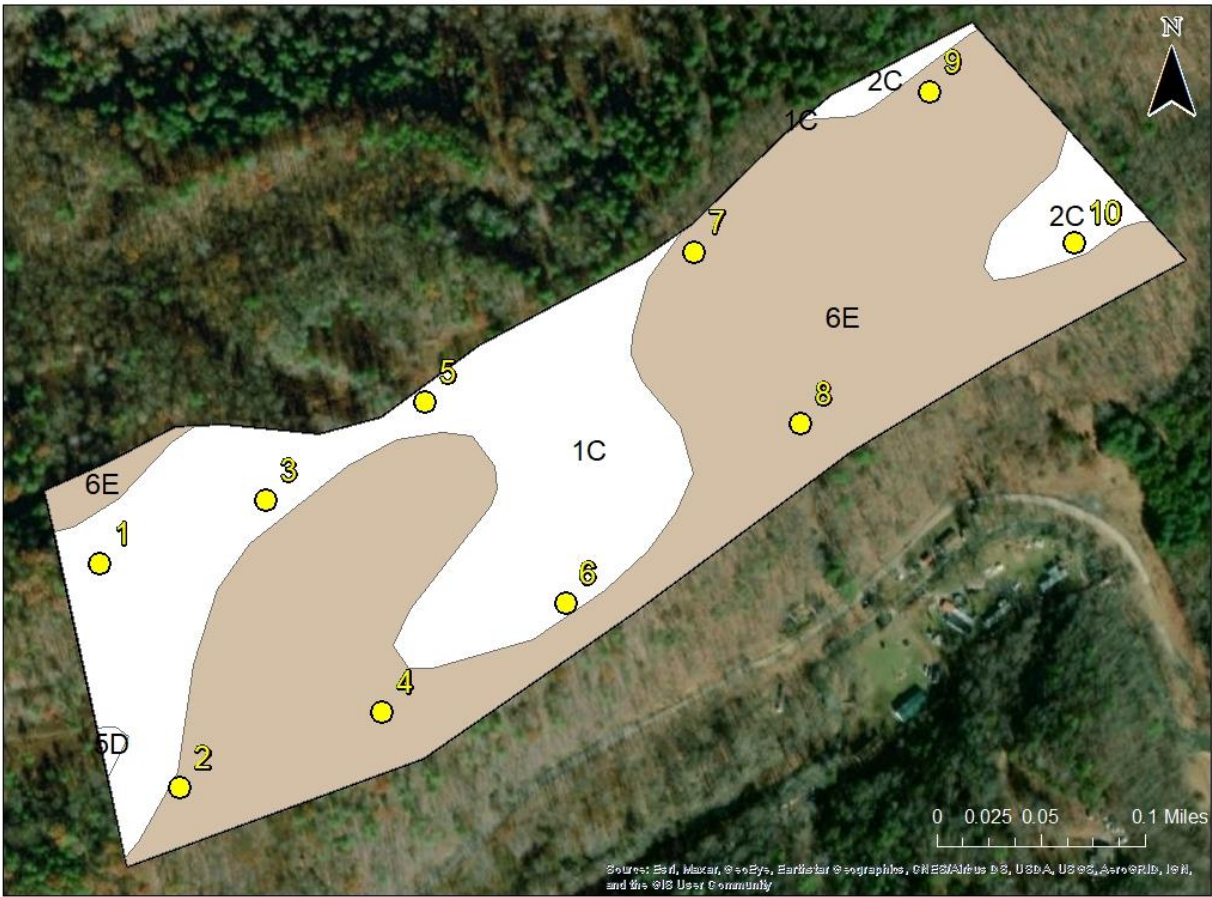
Typical profile

- *H1 - 0 to 4 inches:* very channery silt loam
- *H2 - 4 to 13 inches:* very channery silt loam
- *H3 - 13 to 79 inches:* bedrock

Properties and qualities

- *Slope:* 25 to 65 percent
- *Depth to restrictive feature:* 10 to 20 inches to paralithic bedrock
- *Drainage class:* Somewhat excessively drained
- *Runoff class:* Very high
- *Capacity of the most limiting layer to transmit water (Ksat):* Very low to high (0.00 to 5.95 in/hr)
- *Depth to water table:* More than 80 inches
- *Frequency of flooding:* None
- *Frequency of ponding:* None
- *Available water capacity:* Very low (about 1.0 inches)

VT Fishburn - Soil Map - 6E



Tables — Forest Productivity (Tree Site Index): black oak (Olson 1959 (810)) — Summary By Map Unit

Collapse Summary by Map Unit — Montgomery County, Virginia (VA121)

Summary by Map Unit — Montgomery County, Virginia (VA121)

Map unit symbol	Map unit name	Rating (feet)	Acres in AOI	Percent of AOI
1C	Berks-Clymer complex, 7 to 15 percent slopes	70	11.3	33.1%
2C	Berks-Groseclose complex, 7 to 15 percent slopes	70	1.4	4.2%
5D	Berks-Weikert complex, 15 to 25 percent slopes	70	0.1	0.2%
6E	Berks and Weikert soils, 25 to 65 percent slopes	70	21.4	62.6%
Totals for Area of Interest			34.3	100.0%

Tables — pH (1 to 1 Water) — Summary By Map Unit

Collapse Summary by Map Unit — Montgomery County, Virginia (VA121)

Summary by Map Unit — Montgomery County, Virginia (VA121)

Map unit symbol	Map unit name	Rating	Acres in AOI	Percent of AOI
1C	Berks-Clymer complex, 7 to 15 percent slopes	5.1	11.3	33.1%
2C	Berks-Groseclose complex, 7 to 15 percent slopes	5.1	1.4	4.2%
5D	Berks-Weikert complex, 15 to 25 percent slopes	5.1	0.1	0.2%
6E	Berks and Weikert soils, 25 to 65 percent slopes	5.1	21.4	62.6%
Totals for Area of Interest			34.3	100.0%

Tables — pH (1 to 1 Water) — Summary By Map Unit

Collapse Summary by Map Unit — Montgomery County, Virginia (VA121)

Summary by Map Unit — Montgomery County, Virginia (VA121)

Map unit symbol	Map unit name	Rating	Acres in AOI	Percent of AOI
1C	Berks-Clymer complex, 7 to 15 percent slopes	5.1	11.3	33.1%
2C	Berks-Groseclose complex, 7 to 15 percent slopes	5.1	1.4	4.2%
5D	Berks-Weikert complex, 15 to 25 percent slopes	5.1	0.1	0.2%
6E	Berks and Weikert soils, 25 to 65 percent slopes	5.1	21.4	62.6%
Totals for Area of Interest			34.3	100.0%

VT Fishburn - Topographic Map

